Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Two fixed points A and B are taken on the co-ordinate axes such that $\mathrm{OA}=\mathrm{a}$ and $\mathrm{OB}=\mathrm{b}$. Two variable point A' and B' are taken on the same axes such that $\mathrm{OA}'+\mathrm{OB}'=\mathrm{OA}+\mathrm{OB}$. Find the locus of the point of intersection of AB' and $\mathrm{A}'\mathrm{B}$.

(A) $x-y=a+b$

(B) $x+y=a+b$

(C) $2\mathrm{x}+\mathrm{y}=\mathrm{a}+\mathrm{b}$

(D) $x+3y=a+b$

Q.2

The distance between the two parallel lines is 1 unit. A point A is chosen to lie between the lines at a distance d from one of them. Triangle ABC is equilateral with B on one line and C on other parallel line. The length of the side of the equilateral triangle is-

(A) $(2/3)\sqrt{\mathrm{d}^2+\mathrm{d}+1}$

(B) $2\sqrt{\left(\mathrm{d}^2-\mathrm{d}+1\right)/3}$

(C) $2\sqrt{\mathrm{d}^2-\mathrm{d}+1}$

(D) $\sqrt{\mathrm{d}^2-\mathrm{d}+1}$

Q.3

Find the angle and equation of bisectors of angle between the lines represented by the equation $2x^2-7xy+3y^2=0$

(A) $45^\circ, 7x^2-2xy-7y^2=0$

(B) $30^\circ, 7x^2+2xy-7y^2=0$

(C) $45^\circ, 5x^2-2xy-7y^2=0$

(D) $30^\circ, 7x^2-2xy-5y^2=0$

Q.4

Find the locus of a point which moves such that the sum of the perpendicular distances from it on two given straight lines is a constant.

(A) $\mathrm{mx}-\left(1+\sqrt{1+\mathrm{m}^2}\right)\mathrm{y}=\mathrm{k}\sqrt{1+\mathrm{m}^2}$

(B) $mx+\left(1+\sqrt{1+m^2}\right)y=k\sqrt{1+m^2}$

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(C) $mx-\left(1+\sqrt{1+m^2}\right)y=(-k)\sqrt{1+m^2}$

(D) None of these

Q.5

Let $\mathrm{A_r, r=1,2,3}$ be points on the number line such that $\mathrm{OA_1, OA_2, OA_3, \ldots\ldots}$ are in G.P. where O is the origin and the common ratio of the G.P. be a positive proper fraction. Let $\mathrm{M_r}$ be the middle point of the line segment $\mathrm{A_rA_{r+1}}$. Then the value of $\sum_{r=1}^{\infty}\mathrm{OM_r}$ is equal to -

(A) $\frac{\mathrm{OA_1(OSA_1-OA_2)}}{\mathrm{2(OA_1+OA_2)}}$

(B) $\frac{\mathrm{OA_1(OA_1+OA_2)}}{\mathrm{2(OA_1-OA_2)}}$

(C) $\frac{\mathrm{OA_1}}{\mathrm{2(OA_1-OA_2)}}$

(D) ∞

Q.6

Given $A(0,0)$ and $B(x,y)$ with $x\in(0,1)$ and $y>0$. Let the slope of the line AB equal to m_1. Point C lies on the line $\mathrm{x=1}$ such that the slope of BC equal to $\mathrm{m_2}$ where $\mathrm{0<m_2<m_1}$. If the area of the triangle ABC can be expressed as $\left(m_1-m_2\right)f(x)$, then the largest possible value of $f(x)$ is -

(A) 1

(B) $1/2$

(C) $1/4$

(D) $1/8$

Q.7

If the pairs of lines $x^2+2xy+ay^2=0$ and $ax^2+2xy+y^2=0$ have exactly one line in common, then the joint equation of the other two lines is given by -

(A) $3x^2+8xy-3y^2=0$

(B) $3x^2+10xy+3y^2=0$

(C) $y^2+2xy-3x^2=0$

(D) $x^2+2xy-3y^2=0$

Q.8

A ray of light is incident on x-axis at $A\equiv(-2,0)$ and after reflection from the x-axis it intersect the curve $y^2=4x$ in P and Q. If $\frac{1}{\mathrm{AP}}+\frac{1}{\mathrm{AQ}}=\frac{1}{4}$ ($\mathrm{P,Q}$ being in l-quadrant), then equation of incident ray, is

(A) $x+y+2=0$

(B) $y-x\sqrt{3}-2\sqrt{3}=0$

(C) $y+x\sqrt{3}+2\sqrt{3}=0$

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(D) $x+\sqrt{3}y+2=0$

Q.9

Angles made with x-axis by a straight line drawn through $(1,2)$ so that it intersects $x+y=4$ at a distance $\sqrt{6}/3$ from $(1,2)$ are-

(A) 105°

(B) 75°

(C) 60°

(D) 15°

Q.10

For the straight lines $4x+3y-6=0$ and $5x+12y+9=0$. Choose the correct options -

(A) Bisector of the obtuse angle between them is $9x-7y-41=0$

(B) Bisector of the acute angle between them is $7\mathrm{x}-9\mathrm{y}-3=0$

(C) Bisector of the angle which contains origin is $7\mathrm{x}+9\mathrm{y}-3=0$

(D) Bisector of the angle which contains $(1,2)$ is $9x-7y-41=0$

Q.11

In an acute triangle ABC, if the coordinates of orthocentre ' H ' are $(4,b)$, centroid ' G ' are $(b,2b-8)$ and circumcentre 'S' are $(-4,8)$, then 'b' can not be

(A) 4

(B) 8

(C) 12

(D) -12

Q.12

Passage:
A variable line L is drawn through $O(0,0)$ to meet the lines L_1 and L_2 given by $y-x-10=0$ and $\mathrm{y}-\mathrm{x}-20=0$ at the points A and B, respectively $-$
Question:
A point P is taken on L such that $\frac{2}{OP}=\frac{1}{OA}+\frac{1}{OB}$. Then the locus of P is $-$

(A) $3\mathrm{x}+3\mathrm{y}=40$

(B) $3\mathrm{x}+3\mathrm{y}+40=0$

(C) $3x-3y=40$

(D) $3\mathrm{y}-3\mathrm{x}=40$

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Q.13

Passage:
A variable line L is drawn through $O(0,0)$ to meet the lines L_1 and L_2 given by $y-x-10=0$ and $\mathrm{y}-\mathrm{x}-20=0$ at the points A and B, respectively $-$
Question:
Locus of P, if $\mathrm{OP}^2=\mathrm{OA}\times\mathrm{OB}$, is $-$

(A) $(\mathrm{y}-\mathrm{x})^2=100$

(B) $(\mathrm{y}+\mathrm{x})^2=50$

(C) $(\mathrm{y}-\mathrm{x})^2=200$

(D) None of these

Q.14

Passage:
A variable line L is drawn through $O(0,0)$ to meet the lines L_1 and L_2 given by $y-x-10=0$ and $\mathrm{y}-\mathrm{x}-20=0$ at the points A and B, respectively $-$
Question:
Locus of P, if $\frac{1}{\mathrm{OP}^2}=\frac{1}{\mathrm{OB}^2}+\frac{1}{\mathrm{OA}^2}$, is $-$

(A) $(\mathrm{y}-\mathrm{x})^2=80$

(B) $(y-x)^2=100$

(C) $(\mathrm{y}-\mathrm{x})^2=64$

(D) None of these

Q.15

Passage:
Consider points $A(3,5)$ and $B(7,15)$.
Question:
If P is a point on the line $y=x$ such that $PA+PB$ is minimum, then coordinates of P are $-$

(A) $\left(\frac{31}{5},\frac{31}{5}\right)$

(B) $\left(\frac{27}{5},\frac{27}{5}\right)$

(C) $\left(\frac{13}{5},\frac{13}{5}\right)$

(D) $\left(\frac{12}{5},\frac{12}{5}\right)$

Q.16

Passage:
Consider points $A(3,5)$ and $B(7,15)$.
Question:
Let P be a point on the line $4\mathrm{x}+3\mathrm{y}+7=0$. Then the coordinates of P such that $|\mathrm{PA}-\mathrm{PB}|$ is maximum are
-

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(A) $\left(\frac{1}{7}, \frac{55}{23}\right)$

(B) $\left(\frac{1}{23}, \frac{-55}{23}\right)$

(C) $\left(\frac{3}{7}, \frac{1}{7}\right)$

(D) $\left(\frac{3}{7}, \frac{5}{7}\right)$

Q.17

Passage:

Consider points $A(3, 5)$ and $B(7, 15)$.

Question:

P and Q are points on the line joining A and B such that $AP = PQ = QB$. A square $PQRS$ is constructed on PQ as one side, the coordinates of R can be -

(A) $\left(-5, \frac{29}{3}\right)$

(B) $\left(\frac{7}{3}, 13\right)$

(C) $\left(-\frac{11}{3}, -15\right)$

(D) $\left(-\frac{11}{3}, 15\right)$

Q.18

Match the column -

	Column I	**Column II**
(A)	The value k for which $4x^2 + 8xy + ky^2 = 9$ is the equation of a pair of straight lines is	(p) 3
(B)	If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then the value of c is	(q) –3
(C)	If the gradient of one of the lines $x^2 + hxy + 2y^2 = 0$ is twice that of the other, then h =	(r) 2
(D)	If the lines $ax^2 + 2hxy + by^2 = 0$ are equally inclined to the lines $ax^2 + 2hxy + by^2 + \lambda(x^2 + y^2) = 0$, then the value of λ can be	(s) 4

(A) (A) - p,s, (B) - p,s, (C) - r,s, (D) - q,s

(B) (A) - s, (B) - r, (C) - p,q, (D) - p,q,r,s

(C) (A) - p,r, (B) - r,s, (C) - p,s, (D) - p,q,s

(D) (A) - r, (B) - q,r,s, (C) - q,r, (D) - q,r,s

Q.19

The absolute value of c, so that the lines joining the origin to the common points of $(x - 3)^2 + (y - 4)^2 = c^2$ and $4x + 3y = 24$ are at right angles.

Q.20

If $2a^2 - 7ab - ac + 3b^2 - 2bc - c^2 = 0$ then the family of lines $ax - by + c = 0$ are either concurrent at

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the point $P(x_1, y_1)$ or at the point $Q(x_2, y_2)$ then the distance of the origin from the line through the points P and Q, the distance being measured parallel to the line $3x - 4y = 2$ is

Answers & Solutions

Q.1 Answer:

$x + y = a + b$

Solution:

Let $A \equiv (a, 0), B \equiv (0, b), A' \equiv (a', 0), B' \equiv (0, b')$

Equation of $A'B$ **is** $\frac{x}{a'} + \frac{y}{b} = 1$ **...(1)**

and the equation of AB' **is** $\frac{x}{a} + \frac{y}{b'} = 1$ **...(2)**

Subtracting (1) from (2), we get, $x\left(\frac{1}{a} - \frac{1}{a'}\right) + y\left(\frac{1}{b'} - \frac{1}{b}\right) = 0$

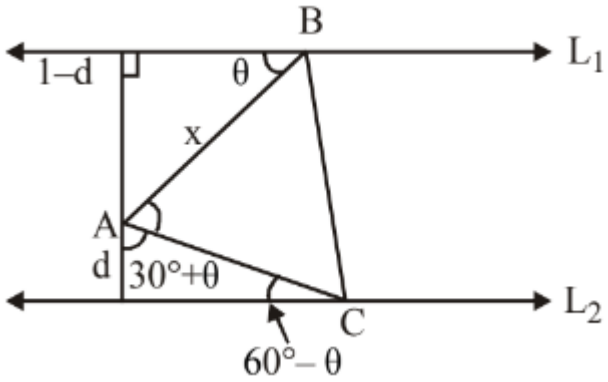
$$\Rightarrow \frac{x(a'-a)}{aa'} + \frac{y(b-b')}{bb'} = 0$$
$$\Rightarrow \frac{x}{aa'} + \frac{y}{bb'} = 0 \quad [\text{Using } a' - a = b - b']$$
$$\Rightarrow b' = \frac{a(a+b)y}{ay - bx} \ldots (3)$$
$$\text{From (2) } b'x + ay = ab' \Rightarrow b' = \frac{ay}{a-x} \ldots (4)$$

Equations (3) and (4) we get $x + y = a + b$**, which is the required locus.**

Q.2 Answer:

$2\sqrt{(d^2 - d + 1)/3}$

Solution:



From the figure,

$x\cos(\theta + 30^\circ) = d$ **.....(1)**

$x\sin\theta = 1 - d$ **..... (2)**

Dividing (1) by (2), we have

$\sqrt{3}\cot\theta = \frac{1+d}{1-d}$

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Squaring eq. (2) and putting the value of $\cot\theta$, we have $x^2 = \frac{1}{3}\left(4d^2 - 4d + 4\right)$

$\Rightarrow \quad x = 2\sqrt{\frac{d^2-d+1}{3}}$

Q.3 Answer:

$45^\circ, 7x^2 - 2xy - 7y^2 = 0$

Solution:

Since $2x^2 - 7xy + 3y^2 = (2x - y)(x - 3y)$

$\therefore$ Given lines are $2x - y = 0$ and $x - 3y = 0$

since $a = 2, h = -7/2, b = 3$

Therefore angle between lines given by $\tan\theta = \frac{2\sqrt{(-7/2)^2-6}}{2+3} = 1 \Rightarrow \theta = 45^\circ$

Again equations of bisectors are given by $\frac{x^2-y^2}{2-3} = \frac{xy}{-7/2} \Rightarrow 7x^2 - 2xy - 7y^2 = 0$

Q.4 Answer:

$\mathrm{mx} - \left(1 + \sqrt{1+\mathrm{m}^2}\right)\mathrm{y} = \mathrm{k}\sqrt{1+\mathrm{m}^2}$

Solution:

We may take one of the two given straight line to be our x-axis and the point of intersection of the given lines as our origin.

Observe that we have the freedom of choosing our axes, depending on the problem.

By our choice of the axes one of the given lines is $y = 0$ and let the other be $mx - y = 0$.

If $\mathrm{P}\left(\mathrm{x}', \mathrm{y}'\right)$ is the variable point, its distances from the two given lines are $|\mathrm{y}'|$ and $\left|\frac{\mathrm{mx}'-\mathrm{y}'}{\sqrt{1+\mathrm{m}^2}}\right|$,

we are given that P moves such that

$|\mathrm{y}'| + \left|\frac{\mathrm{mx}'-\mathrm{y}'}{\sqrt{1+\mathrm{m}^2}}\right| = \mathrm{k} =$ constant, $\mathrm{k} > 0$.

$\therefore$ The locus of p is $(\sqrt{1+\mathrm{m}^2})|\mathrm{y}| + |\mathrm{mx} - \mathrm{y}| = \mathrm{k}\sqrt{1+\mathrm{m}^2}$.

In region I where $\mathrm{y} > 0,\ \mathrm{mx} > \mathrm{y}$ the locus is the straight line $\mathrm{mx} + \sqrt{1+\mathrm{m}^2} \quad \mathrm{y} - \mathrm{k}\sqrt{1+\mathrm{m}^2} = 0$.

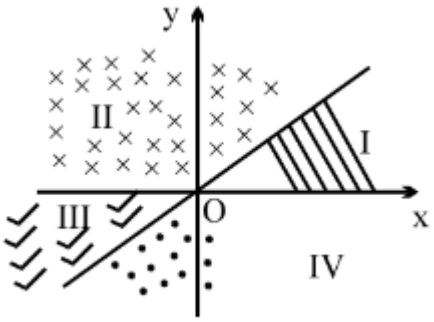
In region II where $y > 0, mx < y$ the locus is the straight line $\left(1 + \sqrt{1+m^2}\right) y - mx = k\sqrt{1+m^2}$.

In region III, where $\mathrm{y} < 0,\ \mathrm{mx} < \mathrm{y}$ the locus is the straight line $\left(1 - \sqrt{1+\mathrm{m}^2}\right)\mathrm{y} - \mathrm{mx} = \mathrm{k}\sqrt{1+\mathrm{m}^2}$.

In region IV, where $\mathrm{y} < 0, \mathrm{mx} > \mathrm{y}$ the locus is

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$mx - \left(1+\sqrt{1+m^2}\right) y = k\sqrt{1+m^2}.$



Q.5 Answer:

$\frac{OA_1(OA_1+OA_2)}{2(OA_1-OA_2)}$

Solution:

$OM_r = OA_r + \frac{OA_{r+1}-OA_r}{2} = \frac{OA_r+OA_{r+1}}{2}$

$= \frac{1}{2}\left\{OA_1 \times k^{r-1} + OA_1 \times k^r\right\} = \frac{OA_1}{2}(1+k)k^{r-1}$

$\sum_{r=1}^{\infty} OM_r = \frac{OA_1}{2}(1+k)\sum_{r=1}^{\infty} k^{r-1} = \frac{OA_1}{2}(1+k) \times \frac{1}{1-k}$

$= \frac{OA_1}{2} \times \frac{OA_2}{1+\frac{OA_1}{1-\frac{OA_2}{OA_1}}} = \frac{OA_1(OA_1+OA_2)}{2(OA_1-OA_2)}$

Q.6 Answer:

$1/8$

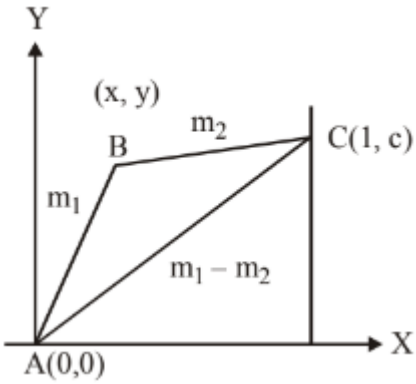
Solution:

Let the coordinates of C be $(1, c)$. Then, $m_2 = \frac{c-y}{1-x}$ or $m_2 = \frac{c-m_1x}{1-x}$

$\Rightarrow m_2 - m_2x = c - m_1x$

$\Rightarrow \quad (m_1 - m_2)\, x = c - m_2$

$\Rightarrow c = (m_1 - m_2)\, x + m_2$ **.....(1)**



Now area $\triangle ABC$ is

$$\frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ x & m_1x & 1 \\ 1 & c & 1 \end{vmatrix} = \frac{1}{2}\left[cx - m_1x\right] = \frac{1}{2}\left|\left[\left((m_1-m_2)\,x + m_2\right)x - m_1x\right]\right|$$

$$= \frac{1}{2}\left|\left[(m_1-m_2)\,x^2 + m_2x - m_1x\right]\right. = \frac{1}{2}(m_1-m_2)\left(x-x^2\right) \left[\because x > x^2 \text{ in } (0,1)\right]$$

Hence, $f(x) = \frac{1}{2}\left(x-x^2\right)$

$f(x)_{\max} = \frac{1}{8}$ when $x = \frac{1}{2}$

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Q.7 Answer:

$3x^2 + 10xy + 3y^2 = 0$

Solution:

Let $y = mx$ be a line common to the given pairs of lines. Then

$am^2 + 2m + 1 = 0$ and $m^2 + 2m + a = 0$

$\Rightarrow \quad \frac{m^2}{2(1-a)} = \frac{m}{a^2-1} = \frac{1}{2(1-a)} \Rightarrow m^2 = 1$ and $m = -\frac{a+1}{2}$

$\Rightarrow (a+1)^2 = 4$

$\Rightarrow \quad a = 1$ or -3

But for $a = 1$, the two pairs have both the lines common.

So, $a = -3$ and the slope m of the line common to both the pairs is 1.

Now, $x^2 + 2xy + ay^2 = x^2 + 2xy - 3y^2 = (x-y)(x+3y)$

and $ax^2 + 2xy + y^2 = -3x^2 + 2xy + y^2 = -(x-y)(3x+y)$

So, the equation of the required lines is $(x+3y)(3x+y) = 0$

$\Rightarrow 3x^2 + 10xy + 3y^2 = 0$

Q.8 Answer:

$y + x\sqrt{3} + 2\sqrt{3} = 0$

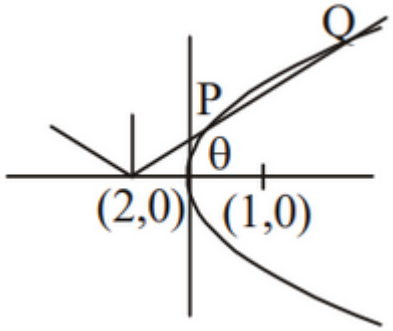
Solution:

Let P is $(-2 + r\cos\theta, 0 + r\sin\theta)$

$\Rightarrow r^2\sin^2\theta - 4(-2 + r\cos\theta) = 0$

$\Rightarrow r^2\sin^2\theta - 4r\cos\theta + 8 = 0$

$\Rightarrow r_1 + r_2 = \frac{4\cos\theta}{\sin^2\theta}$ and $r_1 r_2 = \frac{8}{\sin^2\theta}$



$\frac{1}{AP} + \frac{1}{AQ} = \frac{1}{r_1} + \frac{1}{r_2} = \frac{\cos\theta}{2} = \frac{1}{4}$

$\Rightarrow \tan\theta = \sqrt{3} \Rightarrow$ Slope of incident ray $\tan(\pi - \theta) - \tan\theta$

Hence equation of incident ray $y - 0 = -\sqrt{3}(x+2)$

$\Rightarrow y + x\sqrt{3} + 2\sqrt{3} = 0$

Q.9 Answer:

75°

Solution:

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Let the angle be θ.

Then, equation of the given line is

$\frac{x-1}{\cos\theta} = \frac{y-2}{\sin\theta}$

. . . (1)

The coordinates of a point on (1) at a distance $\sqrt{6}/3$ from $(1, 2)$ are

$(1 + \sqrt{6}/3\cos\theta, 2 + \sqrt{6}/3\sin\theta)$.

This point lies on $x + y = 4$.

Therefore, $1 + \frac{\sqrt{6}}{3}\cos\theta + 2 + \frac{\sqrt{6}}{3}\sin\theta = 4 \Rightarrow \cos\theta + \sin\theta = \sqrt{\frac{3}{2}}$

$\Rightarrow \quad \frac{1}{\sqrt{2}}\cos\theta + \frac{1}{\sqrt{2}}\sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \cos\left(\theta - \frac{\pi}{4}\right) = \cos\left(\pm\frac{\pi}{6}\right) \Rightarrow \theta - \frac{\pi}{4} = \pm\frac{\pi}{6}$

$\Rightarrow \theta = 75^\circ$ **or** $\theta = 15^\circ$

Q.10 Answer:

Bisector of the obtuse angle between them is $9x - 7y - 41 = 0$

Solution:

Equations of bisectors of the angles between the given lines are

$$\frac{4x + 3y - 6}{\sqrt{4^2 + 3^2}} = \pm\frac{5x + 12y + 9}{\sqrt{5^2 + 12^2}} \Rightarrow \frac{4x + 3y - 6}{5} = \pm\frac{5x + 12y + 9}{13}$$

$$\Rightarrow 52x + 39y - 78 = \pm(25x + 60y + 45) \Rightarrow 27x - 21y - 123 = 0, 77x + 99y - 33 = 0$$

$$\Rightarrow 9x - 7y - 41 = 0, 7x + 9y - 3 = 0.$$

Let the angle between the line $4x + 3y - 6 = 0$ and the bisector $9x - 7y - 41 = 0$ be ' θ ', then

$\tan\theta = \left|\frac{-\frac{4}{3} - \frac{9}{7}}{1 + \left(\frac{-4}{3}\right)\frac{9}{7}}\right| = \frac{11}{3} > 1$

Hence, (A) The bisector of the obtuse angle is $9x - 7y - 41 = 0$

(B) The bisector of the acute angle is $7x - 9y - 3 = 0$

(C) The bisector of the angle containing the origin $= \frac{-4x-3y+6}{\sqrt{(-4)^2+(-3)^2}} = \frac{5x+12y+9}{\sqrt{5^2+12^2}}$

$\Rightarrow \quad 7x + 9y - 3 = 0$

(D) For the point $(1, 2)$, $4x + 3y - 6 = 4 \times 1 + 3 \times 2 - 6 > 0$

$5x + 12y + 9 = 12 \times 2 + 9 > 0$

Hence equation of the bisector of the angle containing the point $(1, 2)$ is $\frac{4x+3y-6}{5} = \frac{5x+12y+9}{13} \Rightarrow 9x - 7y - 41 = 0$

Q.11 Answer:

4

Solution:

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As H, G and C are collinear

$$\therefore \begin{vmatrix} 4 & b & 1 \\ b & 2b-8 & 1 \\ -4 & 8 & 1 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 4 & b & 1 \\ b-4 & b-8 & 0 \\ -(b+4) & 16-2b & 0 \end{vmatrix} = 0$$

$$\Rightarrow (b-4)(16-2b)+(b+4)(b-8)=0$$

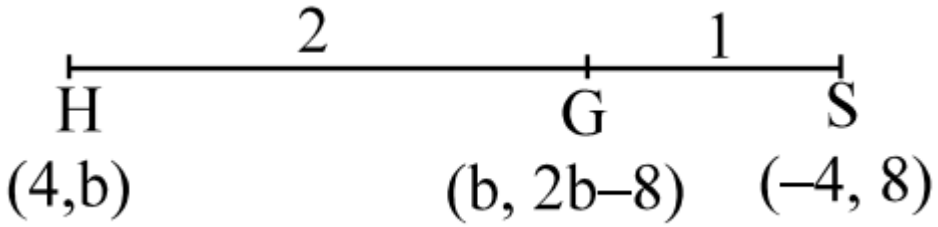
$$\Rightarrow 2(b-4)(8-b)+(b+4)(b-8)=0$$

$$\Rightarrow (8-b)[(2b-8)-(b+4)]=0$$

$$\Rightarrow (8-b)(b-12)=0$$

Hence $b=8$ or 12 , which is wrong because collinearity does not explain centroid, orthocentre and circumcentre.

Now



$$\therefore \quad \frac{-8+4}{3} = b \Rightarrow b = \frac{-4}{3}$$

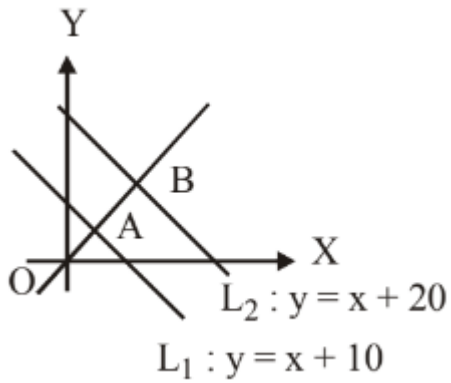
and $\frac{16+b}{3} = 2b-8 \Rightarrow b = 8$

But no common value of ' b ' is possible

Q.12 Answer:

$3y - 3x = 40$

Solution:



(i) Let the parametric equation of drawn line be $\frac{x}{\cos\theta} = \frac{y}{\sin\theta} = r$

$$\Rightarrow \quad x = r\cos\theta, y = r\sin\theta$$

Putting it in L_1, we get

$$r\sin\theta = r\cos\theta + 10$$

$$\Rightarrow \quad \frac{1}{OA} = \frac{\sin\theta - \cos\theta}{10}$$

Similarly putting the general point of drawn line in the equation of L_2 we get

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$$\Rightarrow \frac{1}{OB} = \frac{\sin\theta - \cos\theta}{20}$$

Let $P = (h, k)$ and $OP = r$

$r\cos\theta = h, r\sin\theta = k,$ we have

$\Rightarrow 40 = 3r\sin\theta - 3r\cos\theta \Rightarrow 3y - 3x = 40$

(ii) $r^2 = \frac{10}{\sin\theta - \cos\theta} \times \frac{20}{\sin\theta - \cos\theta} \Rightarrow (r\sin\theta - r\cos\theta)^2 = 200$

Hence, locus is $(y - x)^2 = 200$

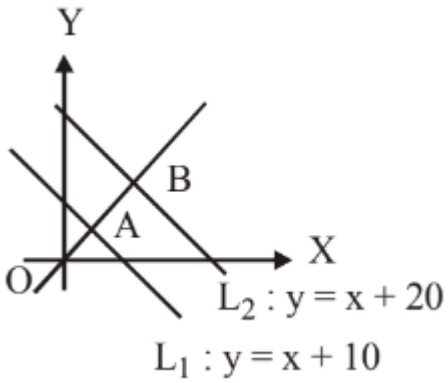
(iii) $\frac{1}{r^2} = \frac{(\sin\theta - \cos\theta)^2}{100} + \frac{(\sin\theta - \cos\theta)^2}{400} \Rightarrow 400 = 5(r\sin\theta - r\cos\theta)^2$

Hence, the locus is $400 = 5(x - y)^2$ **i.e.** $(x - y)^2 = 80$

Q.13 Answer:

$(y - x)^2 = 200$

Solution:



(i) Let the parametric equation of drawn line be $\frac{x}{\cos\theta} = \frac{y}{\sin\theta} = r$

$\Rightarrow \quad x = r\cos\theta, y = r\sin\theta$

Putting it in L_1**, we get**

$$r\sin\theta = r\cos\theta + 10$$

$$\Rightarrow \quad \frac{1}{OA} = \frac{\sin\theta - \cos\theta}{10}$$

Similarly putting the general point of drawn line in the equation of L_2 **we get**

$$\Rightarrow \frac{1}{OB} = \frac{\sin\theta - \cos\theta}{20}$$

Let $P = (h, k)$ and $OP = r$

$r\cos\theta = h, r\sin\theta = k,$ we have

$\Rightarrow 40 = 3r\sin\theta - 3r\cos\theta \Rightarrow 3y - 3x = 40$

(ii) $r^2 = \frac{10}{\sin\theta - \cos\theta} \times \frac{20}{\sin\theta - \cos\theta}$

$\Rightarrow (r\sin\theta - r\cos\theta)^2 = 200$

Hence, locus is $(y - x)^2 = 200$

(iii) $\frac{1}{r^2} = \frac{(\sin\theta - \cos\theta)^2}{100} + \frac{(\sin\theta - \cos\theta)^2}{400}$

$\Rightarrow 400 = 5(r\sin\theta - r\cos\theta)^2$

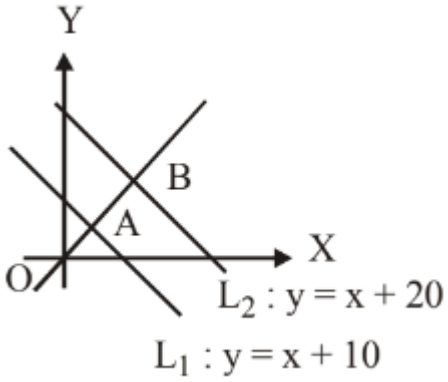
Hence, the locus is $400 = 5(x - y)^2$ **i.e.** $(x - y)^2 = 80$

Q.14 Answer:

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$(y-x)^2 = 80$

Solution:



(i) Let the parametric equation of drawn line be $\frac{x}{\cos\theta} = \frac{y}{\sin\theta} = r$

$\Rightarrow \quad x = r\cos\theta, y = r\sin\theta$

Putting it in L_1**, we get**

$$r\sin\theta = r\cos\theta + 10$$

$$\Rightarrow \quad \frac{1}{OA} = \frac{\sin\theta - \cos\theta}{10}$$

Similarly putting the general point of drawn line in the equation of L_2 **we get**

$$\Rightarrow \frac{1}{OB} = \frac{\sin\theta - \cos\theta}{20}$$

Let $P = (h, k)$ and $OP = r$

$r\cos\theta = h, r\sin\theta = k,$ we have

$\Rightarrow 40 = 3r\sin\theta - 3r\cos\theta \Rightarrow 3y - 3x = 40$

(ii) $r^2 = \frac{10}{\sin\theta - \cos\theta} \times \frac{20}{\sin\theta - \cos\theta} \Rightarrow (r\sin\theta - r\cos\theta)^2 = 200$

Hence, locus is $(y-x)^2 = 200$

(iii) $\frac{1}{r^2} = \frac{(\sin\theta - \cos\theta)^2}{100} + \frac{(\sin\theta - \cos\theta)^2}{400}$

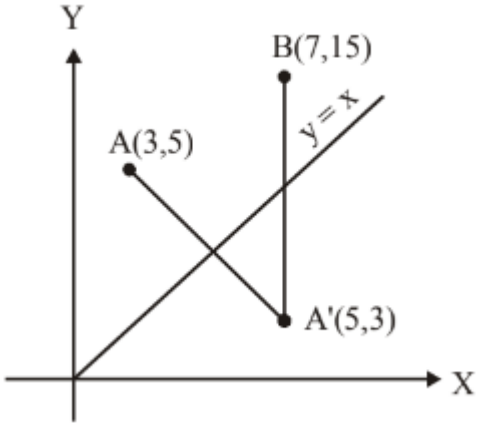
$\Rightarrow 400 = 5(r\sin\theta - r\cos\theta)^2$

Hence, the locus is $400 = 5(x-y)^2$ **i.e.** $(x-y)^2 = 80$

Q.15 Answer:

$\left(\frac{27}{5}, \frac{27}{5}\right)$

Solution:



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(i) Consider a point A', the image of A in $y = x$

$\therefore$ Coordinate of $A' = (5, 3)$

Then $PA = PA'$, thus $PA + PB$ is minimum.

If $PA' + PB$ is minimum, i.e., P, A', B are collinear.

Now, A'B is $y - 3 = \frac{15-3}{7-5}(x - 5) \Rightarrow 2y - 12x + 54 = 0$

It intersects $y = x$ at $\left(\frac{27}{5}, \frac{27}{5}\right)$ which is the required point P.

(ii) Equation of AB is $\frac{y-5}{15-5} = \frac{x-3}{7-3}$

$\Rightarrow \frac{y-5}{10} = \frac{x-3}{4} \Rightarrow \frac{y-5}{5} = \frac{x-3}{2} \Rightarrow 5x - 2y - 5 = 0$

$|PA - PB| \leq AB$, thus $|PA - PB|$ is maximum if points A, B and P are collinear.

Solving $5x - 2y - 5 = 0$ and $4x + 3y + 7 = 0$, the point $P\left(\frac{1}{23}, \frac{-55}{23}\right)$.

(iii) P, Q divides AB in the ratio of $1 : 2$ and $2 : 1$ respectively and hence the coordinates of P and Q are $\left[\frac{1\times7+2\times3}{3}, \frac{1\times15+2\times5}{3}\right]$ and $\left[\frac{2\times7+1\times3}{3}, \frac{1\times5+2\times15}{3}\right]$

$P\left(\frac{13}{3}, \frac{25}{3}\right), Q\left(\frac{17}{3}, \frac{35}{3}\right); r = PQ = \frac{2\sqrt{29}}{3}$

Slope of $PQ = \frac{5}{2}$, Slope of $RQ = -\frac{2}{5} = \tan\theta$

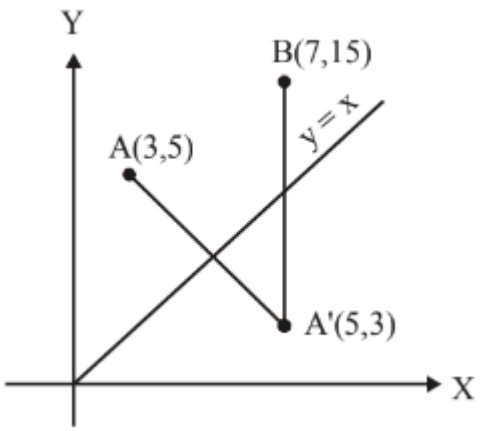
So, $R \equiv \left(\frac{17}{3} \pm \frac{2\sqrt{29}}{3}\left(\frac{-5}{\sqrt{29}}\right), \frac{35}{3} \pm \frac{2\sqrt{29}}{3}\left(\frac{2}{\sqrt{29}}\right)\right)$

R may $\left(\frac{7}{3}, 13\right)$ or $\left(9, \frac{31}{3}\right)$

Q.16 Answer:

$\left(\frac{1}{23}, \frac{-55}{23}\right)$

Solution:



(i) Consider a point A', the image of A in $y = x$

$\therefore$ Coordinate of $A' = (5, 3)$

Then $PA = PA'$, thus $PA + PB$ is minimum.

If $PA' + PB$ is minimum, i.e., P, A', B are collinear.

Now, A'B is $y - 3 = \frac{15-3}{7-5}(x - 5) \Rightarrow 2y - 12x + 54 = 0$

It intersects $y = x$ at $\left(\frac{27}{5}, \frac{27}{5}\right)$ which is the required point P.

(ii) Equation of AB is $\frac{y-5}{15-5} = \frac{x-3}{7-3} \Rightarrow \frac{y-5}{10} = \frac{x-3}{4} \Rightarrow \frac{y-5}{5} = \frac{x-3}{2}$

$\Rightarrow 5x - 2y - 5 = 0$

$|PA - PB| \leq AB$, thus $|PA - PB|$ is maximum if points A, B and P are collinear.

Solving $5x - 2y - 5 = 0$ and $4x + 3y + 7 = 0$, the point $P\left(\frac{1}{23}, \frac{-55}{23}\right)$.

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(iii) P, Q divides AB in the ratio of $1:2$ and $2:1$ respectively and hence the coordinates of P and Q are $\left[\frac{1\times7+2\times3}{3}, \frac{1\times15+2\times5}{3}\right]$ and $\left[\frac{2\times7+1\times3}{3}, \frac{1\times5+2\times15}{3}\right]$

$\mathrm{P}\left(\frac{13}{3}, \frac{25}{3}\right), \mathrm{Q}\left(\frac{17}{3}, \frac{35}{3}\right); \mathrm{r} = \mathrm{PQ} = \frac{2\sqrt{29}}{3}$

Slope of $\mathrm{PQ} = \frac{5}{2}$, Slope of $\mathrm{RQ} = -\frac{2}{5} = \tan\theta$

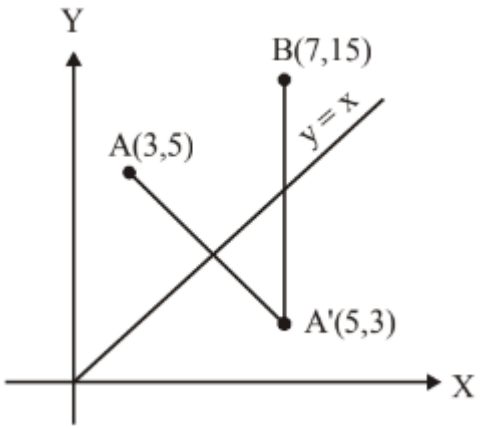
So, $R \equiv \left(\frac{17}{3} \pm \frac{2\sqrt{29}}{3}\left(\frac{-5}{\sqrt{29}}\right), \frac{35}{3} \pm \frac{2\sqrt{29}}{3}\left(\frac{2}{\sqrt{29}}\right)\right)$

R may $\left(\frac{7}{3}, 13\right)$ **or** $\left(9, \frac{31}{3}\right)$

Q.17 Answer:

$\left(\frac{7}{3}, 13\right)$

Solution:



(i) Consider a point A', the image of A in $\mathrm{y} = \mathrm{x}$

$\therefore$ Coordinate of $\mathrm{A}' = (5, 3)$

Then $PA = PA'$, thus $PA + PB$ is minimum.

If $\mathrm{PA}' + \mathrm{PB}$ is minimum, i.e., $\mathrm{P}, \mathrm{A}', \mathrm{B}$ are collinear.

Now, A'B is $y - 3 = \frac{15-3}{7-5}(x-5) \Rightarrow 2y - 12x + 54 = 0$

It intersects $\mathrm{y} = \mathrm{x}$ at $\left(\frac{27}{5}, \frac{27}{5}\right)$ which is the required point P.

(ii) Equation of AB is $\frac{\mathrm{y}-5}{15-5} = \frac{\mathrm{x}-3}{7-3} \Rightarrow \frac{\mathrm{y}-5}{10} = \frac{\mathrm{x}-3}{4}$

$\Rightarrow \frac{\mathrm{y}-5}{5} = \frac{\mathrm{x}-3}{2} \Rightarrow 5\mathrm{x} - 2\mathrm{y} - 5 = 0$

$|\mathrm{PA} - \mathrm{PB}| \leq \mathrm{AB}$, thus $|\mathrm{PA} - \mathrm{PB}|$ is maximum if points A, B and P are collinear.

Solving $5x - 2y - 5 = 0$ and $4x + 3y + 7 = 0$, the point $P\left(\frac{1}{23}, \frac{-55}{23}\right)$.

(iii) P, Q divides AB in the ratio of $1:2$ and $2:1$ respectively and hence the coordinates of P and Q are $\left[\frac{1\times7+2\times3}{3}, \frac{1\times15+2\times5}{3}\right]$ and $\left[\frac{2\times7+1\times3}{3}, \frac{1\times5+2\times15}{3}\right]$

$\mathrm{P}\left(\frac{13}{3}, \frac{25}{3}\right), \mathrm{Q}\left(\frac{17}{3}, \frac{35}{3}\right); \mathrm{r} = \mathrm{PQ} = \frac{2\sqrt{29}}{3}$

Slope of $\mathrm{PQ} = \frac{5}{2}$, Slope of $\mathrm{RQ} = -\frac{2}{5} = \tan\theta$

So, $R \equiv \left(\frac{17}{3} \pm \frac{2\sqrt{29}}{3}\left(\frac{-5}{\sqrt{29}}\right), \frac{35}{3} \pm \frac{2\sqrt{29}}{3}\left(\frac{2}{\sqrt{29}}\right)\right)$

R may $\left(\frac{7}{3}, 13\right)$ **or** $\left(9, \frac{31}{3}\right)$

Q.18 Answer:

(A) - s, (B) - r, (C) - p,q, (D) - p,q,r,s

Solution:

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(A) $-s,(B)-r,(C)-pq,(D)-pqrs$

(A) $4x^2+8xy+ky^2-9=0$ **represents a pair of straight lines if** $h^2-ab=4^2-4k=0\Rightarrow k=4$

(B) $\mathrm{m_1}+\mathrm{m_2}=4\ \mathrm{m_1}\ \mathrm{m_2}$

$\Rightarrow -\frac{2h}{b}=\frac{4a}{b}\Rightarrow -\frac{2(-c)}{-7}=\frac{4\times 1}{-7}\Rightarrow 2c=4\Rightarrow c=2$

(C) Let $\mathrm{m_1},\ \mathrm{m_2}$ **be the slopes of the lines** $\mathrm{x^2+hxy+2y^2=0}$.

Then, $\mathrm{m_1+m_2=-\frac{h}{2},\ m_1\ m_2=\frac{1}{2}}$

But $\mathrm{m_1=2\ m_2}$ **(given).**

Therefore, $\mathrm{3\ m_2=-h/2}$ **and** $\mathrm{2\ m_2^2=1/2}$ **i.e.,** $\mathrm{m_2^2=\frac{1}{4}}$.

Also, $\mathrm{m_2=-h/6}$

$\therefore\quad \mathrm{\frac{h^2}{36}=\frac{1}{4}\Rightarrow h^2=9\Rightarrow h=\pm 3}$

(D) Equation of the bisectors of the angle between the lines $ax^2+2hxy+by^2+\lambda\left(x^2+y^2\right)=0$ **is**

$\frac{x^2-y^2}{(a+\lambda)-(b+\lambda)}=\frac{xy}{h}$ **or** $\frac{x^2-y^2}{a-b}=\frac{xy}{h}$

which is same as the equation of the bisectors of angles between the lines $ax^2+2hxy+by^2=0$.

Thus, the two line pairs are equally inclined to each other for any value of λ.

Q.19 Answer:

—

Solution:

Homogenisation of $x^2+y^2-6x-8y+25-c^2=0\quad \&4x+3y=24$

And taking coefficient of x^2+ **coefficient of** y^2 **as zero**

$\Rightarrow c^2=25$

$c=\pm 5$

$|c|=5$

Q.20 Answer:

—

Solution:

$E=2a^2-7ab-ac+3b^2-2bc-c^2=0$

$\Rightarrow (2a-b+c)(a-3b-c)=0$

$ax+by+c=0$ **is concurrent at** $P(2,-1)$ **or** $Q(-1,3)$

Equation of PQ is $4x+3y=5$

Required distance $=1$

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